

GASTROINTESTINAL SYSTEM

NCLEX STUDY GUIDE



Organized for study and printing | 22 content pages

Gastrointestinal System

Abdominal Examination (I A Pe Pa)

1. Inspection
2. Auscultation
3. Percussion
4. Palpation

Gastroesophageal Reflux Disease GERD

- ☐ Position = reverse Trendelenburg.
- ☐ Back flow of gastric and duodenal content into the esophagus

Barret's Esophagus

- ☐ Pre – Malignant lesion (no es cancer, solo lo predispone).

Board: que lleva follow up: que la RN le dijo a la Pt que tiene un cancer de esophagus bc have a Barret's lesion

Assessment

1. **Heartburn women with heartburn puede ser infarto**

Note: para el board se prioriza porque puede entenderse como un-MI, unless que te especifique que es un-GERD.

2. **Epigastric pain**

Note: Este también se prioriza por la misma razón.

3. Dyspepsia (= indigestion).
4. Regurgitation
5. Hypersalivation
6. Dental Caries
7. Esophagitis
8. **Difficulty to swallowing** (bc stenosis complication, risk for aspiration) = **este es el unico que se reporta.**

Note: Pain que se priorizan

1. **Chest Pain (heartburn)**

2. **Epigastric pain can be:**

- Infarto,
- Pre-eclampsia eclampsia,
- Addisonian crisis,
- **Diabetes ketoacidosis.**

3. **Lower back Pain can be:**

- Acute Hemolytic transfusion reaction
- abdominal aneurysm rupture.

4. **The worse headache of my life can be:**

- cerebral aneurysm rupture
- sub-arachnoidian hemorrhage.

Risk Factors for GERD

1. Caffeinate Beverages
2. Chocolate
3. Alcohol
4. Citrus Fruits
5. Tomatoes and tomatoes products
6. Smoking
7. Peppermint
8. Nasogastric tube replacement

9. Medications

- Anticholinergics (atropine, **dicyclomine**) este es el de Eleccion para-GI
- Theophylline
- Calcium channel blockers (diltiazem, verapamil, nifedipine)
- Nitratos (nitroglycerine)

Intervention

1. Avoid risk factor.
2. Instruct client to eat low fat, high fiber diet.
 - ☐ **Avoid drinking 2 hours before bedtime.**
3. Avoid tight clothes.
- 4.

Elevated HOB 6-8 inches blocks during nights or position in reverse trendelenburg.

5. Avoid Anticholinergic
 - ☐ Avoid Nsaid, ASA (bleeding risk)

6. Medication to Treat GERD

- **Anti Histaminic 2 at bedtime**
 - ☐ Cimetidine, ranitidine
- **Metoclopramide (choice- 30 mint before meals)**
- ☐ Increased gastric empty.
- ☐ Extrapyramidal side effect
- ☐ **Contraindicated:**
 - o Pregnant
 - o Children less than 2 years
 - o Dumping Syndrome

Nota: extrapyramidal effect = Parkinson effects.

Contraindication

Dicyclomine

Gastritis

☐ Inflammation of the stomach or gastric mucosa.

Risk Factor

- Helicobacter Pylori (#1 cause) - Highly seasoned food. = es una bacteria
- ASA - NSAID - Alcohol
- Radiation Therapy – Smoke - GERD

Assessment	
Acute	Chronic
- Abdominal pain Anorexia. - Nausea, vomiting - Hematemesis - Asociada con algo que comistes o tomastes	1. Anorexia, nausea, vomiting 2. Belching/burping (eruptos) 3. Heartburn after eating. 4. Sour taste in the mouth 5. Pernicious anemia Board: chronic gastritis te lo relacionan con pernicious anemia = Vit B12 deficiency. - Se inyectan long life B12 - P. A. = Ataxia (paresthesia) - S/S = Red beefy tongue (shiny) - Fatigue, pallor. Ataxia

Diagnostic

1. Esophagus- gastric- duodenoscopy + biopsy.
2. Urea breath test (H. Pylori confirmatory)

The client consumes a capsule of carbon urea and provide un breath sample 10-20 minutes later.

Para hacer ese exam Avoid antibiotic bismuth subsalicylate (Pepto-Bismol) for 1 month before. Avoid omeprazole and sucralfate= 1 week before

Peptic Ulcer Disease

☐ Ulceration in the mucosal wall of the stomach, pylorus, duodenum or esophagus, erosion may extend through the muscle.

Complication = perforation.

Risk factors

- Same to gastritis
- Helicobacter Pylori (#1 cause)
- Highly seasoned food.
- ASA - NSAID
- Alcohol
- Radiation Therapy
- Smoke
- GERD

Assessment	
Gastric	Duodenal
- Gnawing, sharp pain in the left of the mid epigastrium. - It occurs 30-60 min after a meal. - Food ingestion increased pain. Board: cual es el gastric? Hematemesis is more common than Melena	- Burning pain - It occurs in the mid epigastric area. - 1 ½ -3 hours after meals. - Often awakens the client during night with pain. - More severe pain with empty stomach. - Melena is more common than hematemesis. - Board: Este te despierta en la noche.

Primero duele el gastric y luego el duodenal

Complication

1. Hemorrhage
 - ☐ Hypotension + Tachycardia
2. Perforation
 - ☐ Board like abdomen pain radiated to the left shoulder.

Diagnostic

-Same as gastritis

Treatment for the 3 (Gastritis, GERD, Ulcer)

Medications

1. Antiacids

1. Aluminum hydroxide (constipation, increased fiber) -
2. Magnesium Hydroxide (diarrhea)
3. Calcium carbonate (tums) (constipation). -

All = Nausea, vomiting

Note: Aluminum hydroxide = Phosphate Depletion (Phosphate Binding) Use in patient with Chronic Kidney injury or high PO₄, you give this Aluminum

Hydroxide to ↓ PO₄, with food or snack

- ☐ Sevelamer
- ☐ Renagel
- ☐ Amphogel

For ulcers or gastritis se da alejado de las comidas = 1 hour before and 1-2 hours after meals.

2. Anti H2 (Histamine receptors site agonist)

1. **Cimetidine:** contraindicate in elderly bc confusion, LOC Alteration.
 - High interaction with medication
2. **Ranitidine**
3. **Famotidine**

Take at bedtime

It's expected during the 1st week after treatment anticholinergic side effect.

3. Cytoprotective Agent (Calfate...)

(Son para proteger el estomago) = Crean una capa en el estomago para que cuando coman la comida no te haga mal.

Example= Sucralfate.

Board: take 30 minutes – 1 hour before meals.

4. Proton-pump Inhibitors (Prazoles)

(Mnemonic to know what give with food) Yo **Lanzo** la pelota y **eso** me da gusto, **Lanzo** y **eso** – take with food

1. Lansoprazole with
2. Esomeprazole Food
3. Pantoprazole Take in the morning
4. Omeprazole

Side Effect

- Osteoporosis
- C. Diff Infection (antibiotic + prazoles)
- Kidney Stone.

5. Prostaglandin Analogs

1. Misoprostol: la usan para abortar bc side effect que es miscarriage
takes 30 mint – 1 hour before meals.

Side Effect

- Abdominal pain
- Diarrhea
- Chills
- Miscarriage (pregnant test before)
- Notify HCP if diarrhea persistent more than 1 week or black tarry stool

H. pylory treatment.**Triple Therapy**

Esomeprazole + Amoxicillin + Clarithromycin

Quadruple therapy

-Esomeprazole + Metronidazole + Tetracycline +
Bismuth subsalicylate

-Ranitidine + Metronidazole + Tetracycline + Bismuth
subsalsicylate

*Antibiotic + Prazole + Bismuth

- Surgical Intervention

1.	Billroth I: Partial gastrectomy: ☐ Gastroduodenostomy (<u>anastomosed</u> stomach + duodenum)
2.	Billroth II: Gastrojejunostomy (Anastomosed stomach + jejunum)
3.	Gastric Resection (quitar el estomago completo)

Complication of Billroth I, II, Gastrectomy and bariatric Surgery

Short Term - less than 6 months	Long Term after 6 months
Dumping Syndrome Rapid emptying of gastric content into the small intestine	Pernicious Anemia Vit B12 Deficiency
Assessment	Assessment
Symptoms occurs 30 min after eating - Nausea and vomiting - Abdominal cramping - Borborygmi (peitos en stomago) - Diarrhea / Dizziness - Perspiration - Weakness - Fatigue - Tachycardia	• Red beefy tongue (shiny) • Fatigue • Pallor • Ataxia (paresthesia) Treatment: Vit B12 injected life long

Intervention (Dumping)

- Client education to prevent Dumping syndrome
- Avoid sugar
- Avoid fruit, milk and honey
- Avoid carbohydrates
- Eat Hight protein food
- Eat a Hight FAT diet
- Lie down at least 30 min after meals
- Eat at low fowler position 15
- Avoid Fluid with meals
- Eat Small Frequent food

Medications

- Choice = Dicyclomine

Contraindicate

- Metoclopramide

Hepatitis

All hepatitis are standard precaution expect hepatitis A with bowel incontinence or diarrhea

- Las vocals A and E are = oral and fecal transmission
- Las B, C, D are = parenteral (Blood, body fluids...)

Hepatitis A

- Transmission Fecal oral.
- Contaminated fruits and vegetables, uncooked shellfish.
- Contaminated water, milk
- Poorly washed utensils

Incubation Period: 2 – 6 weeks

Diagnostic

1. Anti HAV in the blood positive
2. IGM increased (immunoglobulins).

Intervention**Prevention**

- Hand washing
- Stool and needle precaution
- Treatment of municipal water supply
- Serologic screening of food handler
- Hep. A vaccine – two doses are need at least 6 months apart
- (children= 1st dose at 12 months but is optional)
- Immunoglobulins for Hep A = recent contact with Hep A who never received Hep A vaccination or had Hep A diseases.

Hepatitis B

Board: Individual at increased risk?

- IV drug abuser
- Long Term Hemodialysis
- Health care worker

Transmission

1. Blood and body fluids.
2. Infected saliva and semen (sexual contact).
3. Contaminated needle.
4. Perinatal and blood or body fluid contact at birth.

Incubation period = 6 – 24 weeks.

Testing

- HBS Ag (antigen de superficies)
- If HBS Ag Positive you have Hep B
- If positive for more than 6 months = Chronic Hep B or carrier.

Board: A toda mujer embarazada hay que hacerle el HBs Ag = if mother antigen is positive when the baby born if weight + 5 pounds = in one vastus lateralis apply Hep B Vaccine + immunoglobulin in the other vastus lateralis.

Now if a baby weighs less than 5 pounds just administer the immunoglobulins.

If antigen de superficies is negative jus apply Hep B if baby are more than 5 pounds

Complications Hep B

- Fulminate hepatitis
- Chronic liver diseases
- Liver Cirrhosis
- Liver Cancer

Intervention**Prevention**

- Hand washing
- Screening blood donors
- Testing all pregnant women
- Needle precaution
- Lead poisoning = 1978 kids (plomo)
 - Board: Hep C, B before 1990 more risk for blood transfusion transmission.
- Avoid sexual contact unprotected and contact with body fluid while HBS Ag = Positive.
- Hep B vaccine
- Hep B vaccine and immunoglobulin if recent contact, and client who never received Hep B vaccine or had Hep B disease.

Note: the only Hepatitis que llevan vaccine are: = A and B

Hepatitis C

Is common among IV Drugs user

Board: Is the major cause of post blood transfusion before 1990

Board: Individual at risk - IV drug abuser

- Health care workers
- Receive frequently transfusion or hemodialysis.

Testing

- Anti HCV positive in blood.

Treatment

- Can use Ribavirin (tablets) = use to in children with bronchiolitis in nebulization and if nurse are pregnant no lo puede dar bc teratogenic and if have contact lenses need to be removed bc precipitating (los cristaliza)

Hepatitis D

Board: Mediterranean and middle Eastern areas.

Hepatitis D occur with Hep B and cause infection only in the presence of HB infection

Testing

- Anti HDV in blood
- No treatment, no vaccination available.

Hepatitis E

- It is a **waterborne** virus.
- Prevalent in areas where sewage disposal is inadequate or where communal bathing in contaminated rivers is practice.

Individual at risk

- Travelers to countries with high risk.

Example: Mexico, Afghanistan, Algeria, India.

Testing

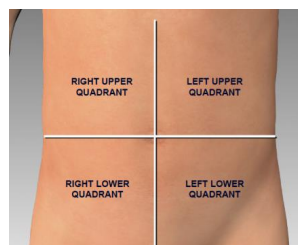
- Anti **HEV** positive
- Transmission: same to A but more water

Complications

Board: **High mortality** rate in **pregnant** women and fetal demise

Assessment Hepatitis

Pre-icteric Stage	Icteric Stage	Post-icteric Stage
- Flu like symptoms - Jaundice. – Malaise - Fatigue. – Abdominal pain - Anorexia. – Headache - Nausea - Muscle ache - Vomiting - Diarrhea	Call Dr if - Jaundice - Clay Feces - Dark Urine - Bc High liver enzymes - Increased bilirubin - Pruritus	- Convalescent stage - Jaundice decreased - The color of urine and stool return to normal. - Pain subsides - Bilirubin and liver enzymes to normal level.
- Increased bilirubin and liver enzymes.		

Abdomen Agudos**Pains**

Right upper quadrant: Gallbladder = **(Cholecystic)** and Liver

Left Upper quadrant: Spleen = board like abdomen radiate and left shoulder.

Mid Epigastrium pain: radiate to the back = **Pancreatitis**

Right lowe quadrant: = **Appendicitis**

Left lower quadrant: = **Diverticulitis**

Note: Flank pain = kidney

Appendicitis**Ubicado en el Cecum**

☐ Occur when the appendix become trapped with hard material (usually feces) that lead to a bacterial infection.

Board: seen more often 10-30 years (teens)

Assessment

1. Client report cramping and pain around the umbilicus and in the mid of epigastric area
 2. Anorexia/nausea and vomiting
 3. As the condition progresses the pain moves to RLQ (Mc Burney's Point) Halfway between the umbilicus and anterior iliac spine. (Press, if pain = Point Mc Positive)
 4. Rebound Tenderness
- At the Mc Burney's point = pain after deep pressure is applied and released.
5. Fever

Board: Appendicitis se llama Appendices Cecal. En que parte de la barriga esta? = En el cecum

Signs of Appendicitis

Dunphy	Rovsing	Iliopsoas	Obturator
Increase pain with coughing	If LLQ is palpate, Pt refers pain In RLQ= referido	Pain during Extension of right hip	Pain with internal rotation Of the right hip or flexion

Complication Peritonitis

- Sudden relieved of pain (perforation)
- Board like abdomen pain radiated to the left shoulder.
- Call the Dr.

Diagnostic

- WBC
- CT Scan = Confirmatory Treatment

Treatment**Surgical Intervention****Pre-Operative**

1. NPO Status
2. IV Fluids
3. Semi fowler Position
4. Apply ICE packs to the abdomen for 20-30 min every jour as prescribe (never heat)
5. Antibiotic
6. Avoid Enemas, laxatives

Post – Operative

1. Monitor sign of infection

Redness at surgical, hot or with discharge (call Dr)

2. NPO Status until bowel sound return
(normal = 72 hours)
3. Morphine for pain
4. Antibiotic

Note: Post- appendicitis & Hernia repair = LPN Care.

Pancreatitis (emergency)

- Autodigestion of the pancreas from premature activation of pancreatic digestive enzymes.
- Acute or chronic
- Associate with large ingestion if fatty food and alcohol

Two primary causes of Pancreatitis

1. Alcohol
2. Biliary tract diseases (gall Stones)

Triggering Factors

- Hyperlipidemia (peptic ulcer disease, High cholesterol, High calcium)
- Alcohol
- Hypercalcemia
- Hyperparathyroidism
- Peptic Ulcer Disease
- Trauma
- After intake of large amount of FAT and alcohol

Assessment

Acute	Chronic
- Abdominal Pain, worsening when lying down or while eating. (Sudden onset at the mid epigastric area radiated to the back.	- Abdominal Pain
Pain relieve a little in fetal position. Never client with pancreatitis in supine position Abdominal tenderness Nausea and vomiting/ weight loss - ↑ WBC ↑ bilirubin ↑ Glucose <u>Signs</u> Cullen's sign = Bluish periumbilical discoloration. Turner's Sign = Flank Hematoma. (Donde unico are expective.	LUQ mass Steatorrhea= fatty stools (grasa y peste). Weight loss Muscles wasting Jaundice Sign of Diabetes

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Complication (Pancreatitis) Call Dr

- Hypocalcemia
- Chvostek
- Trousseaus = TETANY
- Paresthesia's
- Paralytic Ileus
- Ascites
- Hyperglycemia

Diagnostic

- Amylase = Acute Pancreatitis = 500-700
- Lipase = Most Accurate lasting elevate 15 das
- CT Scan = with contrast (more reliable)

Nursing Intervention

P	Pain medication= Morphine if not history of gallbladder (Stone)
A	Antispasmodic = Dicyclomine
N	NPO, NGT for suction, TPN nutrition, no peg tube bc is Enteral nutrition and is contraindicate
C	Calcium
R	Replace fluids and electrolyte
E	Endocrine enzymes (same in cystic fibrosis) = before food Pancreatic enzymes – Pancreatin- Pancrealipasa + Insulin.
A	Antibiotic
S	Steroids

Cholecystitis (RUQ)

□ Inflammation of gallbladder

Risk Factor (F disease)

- FAT – BMI + 30
- Female
- Fertile (1 or more children)
- Forty age (40)
- Genetic predisposition
- Pregnancy

Assessment

1. Nausea, vomiting, indigestion, belching, flatulence
2. Epigastric or RUQ pain that irradiates to the scapula (right shoulder) = (no es priority) = (expective) 2 – 4 hours after sudden eating fatty foods.
3. **Murphy's sign**: firmly placing a hand at the costal margin in the right upper abdominal quadrant and asking the patient to breathe deeply and can't take a deep breath because of the pain

4. Rebound tenderness in the RUQ
5. Fever, tachycardia
6. Sign of dehydration
7. **In case of chronic Cholecystitis**

- Clay Feces
- Jaundice
- Dark Urine
- Pruritus

Note: normal jaundice

- After 24 h after birth =physiologic: Tx. =Sun exposure
- Fluorescing test: contrast that injected to diagnostic diabetes retinopathic.

Diagnostic Procedures

- Ultrasound
- Abdominal X- Ray

Nursing Intervention

1. Maintain NPO during nausea and vomiting episode.
 2. Nasogastric tube for decompression.
 3. IV Fluid
 4. Drug Therapy
- **Meperidine for acute pain**
 - Dicyclomine
 - Antiemetic= Promethazine suppository

Contraindicate

- morphine

Note: Siclemia is opposite to cholecystitis

In siclemia Give morphine and contraindicate Meperidine

Antiemetic (promethazine suppositories)

Surgical Intervention

Open Approach (barriba abierta)	Laparoscopic Approach
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- Cholecystectomy Required 2–3-day hospitalization - T Tube may be placed in the common bile duct. <u>Care of T Tube –</u> Drainage in T for restoration of the Billis to prevent obstruction, with a collector bag (por debajo del drenaje) Monitor and record drainage. - Initially bloody green/yellow, brown = expected during 1st 24 hours, may drain 500 – 600 ml of green - yellow. - If not drainage, or more than 1000 – Call the Dr. - Drainage (via gravity) collection bag, bellow gall bladder level. - Bag empty every 8 hours (OJO). - Board: Assess or check when client is ready to remove T Tube. 1. Feces normal color. 2. Clamp T Tube 1 – 2 before and after meals. If client tolerated food without pain, is ready to removal T – Tube. - Activities Precaution for 4 – 6 weeks - Take shower instead of bath until T- Tube is removed. - Maintain low fat diet (reduce dairy and avoid fried food) - Promote weight reduction. - Avoid gas forming food. - Small and frequently food.	1. Discharge within 24 hours. 2. Educate client regarding free air pain (under right clavicle, shoulder, and scapula) = Normal. 3. Ambulation to expulse gas. Activities are reassured in 1 week.
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Liver Cirrhosis

Chronic progressive liver condition. (Destroyed liver)

Types of Liver cirrhosis

1. Laennec 's (caused by long term alcohol abuse) Board
2. Post Necrotic (due to hepatitis B, C, D)
3. Post hepatic (due to hepatitis B, C, D)
4. Biliary due to bile flow obstruction (hay que quitar la vesicula cuando hay calculos)
5. Cardiac (due to CHF OF THE RIGHT SIDE)

Assessment

Early Sign (expected)

1. Liver enlargement
2. Pain RUQ
3. Weight loss
4. Weakness

5. Fatigue
6. Anorexia
7. Diarrhea
8. Constipation

DERMATOLOGIC FINDINGS

- Caput medusae (dilated abnormal veins) (Como medusa en la barriga)
- Petechia
- Increase skin pigmentation
- Jaundice
- Palmar erythema
- Spider Angiomas

ENDOCRINO FINDINGS

- gynecomastia (= expected, no es priority)

IMMUNE SYSTEM

- Decrease WBC (risk for infection)

NEUROLOGIC FINDING

- **Complication = Ammoniacal Encephalopathy** = (ammonia en el brain) Call the Dr.

Early Sign

1. Confusion
2. LOC Alteration
3. Difficult to arouse (lethargy)
4. **Asterixis** (intentional hand tremors, (que levante las manos y es que tiembla) or liver flap, hand flap) = Writing difficult.

Unintentional hand tremors = Parkinson

Note: liver cirrhosis que no coordina escritura = Call Dr.

Nota: Las bacterias del intestino producen amonía y se aumenta su producción en la cirrosis, causa sangramiento por tanto el riñon no las puede eliminar

Late sign

- Seizure
- coma

Treatment

1. **Decreased protein diet** (bc la ammonia se produce de las proteínas que comemos)
2. Lactulose – soft feces **if diarrhea call Dr bc is a side effect**
3. **Neomycin** (aminoglycoside, se usa también en el GI) **S/E nephrotoxicity, ototoxicity respetan el liver**
- Note: Use neomycin to prevent encephalopathy also in liver biopsy = transhepatic (entrar al higado) to prevent infection. (ascites)
- Transhepatic has risk for bleeding and risk for ammoniacal encephalopathy. Na + Cl van contraries al K

Pulmonary finding

1. SOB
2. Hypoxemia
3. Hyperventilation
4. Ascitis
5. Hypocalcemia (risk for tetany)
6. Hypomagnesemia
7. Peripheral Edema
8. Decreased K = diuretic for choice
 - Spironolactone
 - Triamterene
 - Amiloride

Hematology finding

1. Anemia
2. Coagulation disorder (bc el hígado no metaboliza la vitamina K)
3. Thrombocitopenia
4. Splenomegaly
5. DIC complication: keyword (oozing of blood through IV line) Board

Complication:

1. Portal hypertension
 - Esophageal varices
 - Hemorrhoidal

Esophageal varices

- Dilated and tortuous veins in the esophageal mucosa of esophagus.

Assessment

1. Bleeding (dos tipos de bleeding)
 - Melena
 - Hematemesis = (vómitos con sangre)
2. Jaundice
3. Hepatomegaly, splenomegaly
4. Sign of shock Complication (↓BP+ ↑tachycardia = shock)

Treatment **OJO**

1. 1-Sengstacke – Blakemore tube
 - Use intermittent (infla & desinfla) no puede estar inflada todo el tiempo bc necrose la varice
 - 3 lumen balloon that you need to label
 - Scissors at bed site = bc in case of respiratory difficulty need to cut gastric balloon

Bc el tubo puede hacer un displacement para el stomach y se ahoga el client. Se corta el gástrico para desinflar el balloon.

Complication**1. Hepato-renal syndrome**

Progressive renal failure associated with hepatic failure

- If Pt have a Sudden decrease of urinary output = call Dr.
- Or Increased Creatinine and BUN = Call Dr

Intervention cirrhosis hepatic

1. **Diet:** Protein restriction except for ascites that is edema and need to increase (nuts)
2. Fat soluble vitamins ADEK (same in celiac disease Crohn disease cystic fibrosis)
3. B-blockers (LOL) to decrease BP (nadolol)
4. Iron and folic acid
5. Vit K for bleeding
6. **Oxazepam for anxiety** (todos los benzo se metabolizan en el hígado excepto este)
7. Avoid alcohol and hepatotoxic medication (Tylenol) Acetaminophen (nunca en liver problem)

Ulcerative colitis

- Inflammatory disease of the bowel, mainly the **colon** and **rectum sigmoid** is affected
- Periods of remission and exacerbation

Nota: Crohn disease have 4-6 soft feces with mucus of flemas but NOT blood, in any part desde la boca hasta el ano, una parte mal y una buena casi nunca se opera.

Assessment

1. Severe diarrhea 16-20 loses stool with blood and pus per day
2. More risk for cancer than Crohn disease
3. Tenesmus (deseo de ir al baño todo el tiempo)
4. Anorexia, weight loss, fatigue, malaise
5. Abdominal tenderness and cramping.
6. Dehydration, malnutrition, electrolyte and Acid – Base imbalances (Metabolic acidosis). **K High**
7. Anemia + vit K deficiency.

Diet: Not Milk bc diarrhea

Complication

- Toxic megacolon
- Hemorrhage
- Abscess.
- **More risk for colon cancer than Chorn's disease**

Chron's Disease

- Chronic inflammatory that occurs at **any part** of the intestine. (No cure)
- Most often affects the terminal Ileum and leads to thickening and scarring.
- Skip **lesion on the X-ray**
- Cobblestone appearance (piedritas)
- Severe malnutrition

Assessment

1. Fever
2. Cramp like colicky pain after meals
3. Diarrhea- semi-solid or soft stool with mucus and pus 4-6 x days
4. Steatorrhea (Fatty stools)
5. Abdominal distention
6. Anorexia, nausea, vomiting, weight loss, anemia, malnutrition, electrolytes imbalances, dehydration

Note: Arriba pillory= alkalosis, abajo pillory = acidosis (culito = Acidosis). Intestino delgado donde se absorben los nutrients, Chron's disease

Complication

- Fistula
- Ulceration
- Abscesses

Intervention In Acute setting from Chron and Ulcerative colitis

1. Maintaining NPO Status and IV fluids.
2. TPN
3. Administer anti acids
4. Restrict activities = Bed rest
5. Diet:
 - Low fiber, high protein, and Iron
 - No milk or dairy
 - Avoid gas forming food: Avocado, broccoli, beans, green leafy, vegetable
6. Avoid smoking, alcohol, and caffeine

Treatment = Medication**1. Antibiotic**

- Metronidazole = no with alcohol
- Ciprofloxacin = S/E Achilles tendon rupture= heal pain

Se usan tambien las cefalospirinas (ce) pero se usan mas en cirugias abdominals ex: intestine

2. Aminoacidic salicylates

- Sulfasalazine = Contraindicate in sulfa allergies clients.

Sulfa allergy

- Asa
- Furosemide
- Thiazide diuretic (hydrochlorothiazide)
- Alopurinol
- Gli de diabetes
- Acetazolamide
- Celecoxib autoinflammatory no esterideo, (es sulfa)

- Trimetropin sulfametazole
- Glucose 6 phosphate dehydrogenase = anemia en childrens donde no metabolizan la sulfa

Nota: que tiene Un patient con crohn y colitis = high risk for infection bc las medicinas decrease WBC

Sulfa side effects

- Steven jonhson Syndrome) (rash with skin burn)
antiseizure med son los que mas lo dan

3. Prednisone

Decrease WBC

Healing retardation

4. Immunosuppressant = S/E decrease WBC = sore through

- Azathioprine
- Cyclosporine
- 5. Immunomodulators
- Adalimumab
- Infliximab

Side Effects

- ↓WBC = Risk for infection
- No with life attenuated virus vaccine
- No in active infection
- Risk for Cancer= leukemia
- Anaphylactic reaction (1st dose under supervision) = Sub every other week

Surgical intervention

- Colonostomy and ileostomy

Difference between ileostomy and colostomy

1. El colon se divide en tres partes

- ascending = no se irriga
- traverse = no se irriga
- descending or sigmoid =se irriga

Que se puede irrigar

- el tercio distal del traverse
- descending colostomy

Que no se puede irrigar

- Ileostomy
- Ascending colostomy
- Primera parte del transverse

Note: **la ileostomy nunca se irriga porque la caca es liquida.** Que lleva follow up que irrigaton una ascending colostomy, si dice solo colon si se irriga

Como se le quita el mal olor a la ileostomy: com que comida

- Buttermilk
- Parsley
- Yogurt

1. Ileostomy

- All portion of the large intestine are removed.
- Stoma is in RLQ
- Drainage = liquid – never irrigation
- Risk for dehydration, electrolyte imbalances, skin breakdown.

2. Ascending Colostomy

- Stoma RUQ
- Feces are liquid = risk for breakdown
- No irrigation

3. Sigmoid or descent colostomy

- The distal portion of the colon is removed.
- Stoma = LLQ
- Feces are formed = Irrigation
- Ostomy appliance may eventually not be needed.

Pre-operative nursing intervention

1. Antibiotic before surgery
Ceftriaxone/cephalosporin
2. Diet: High protein, high in carbohydrates, high calories, low residue/fiber= 1 week before surgery
3. Full liquid diet. 24- 48 hours before
4. NPO after midnight.
5. Enemas evening before and the same morning of surgery. If return cloudy don't do it

Post- operative nursing intervention

1. NPO with NG tube until bowel sounds return.
2. Clear liquid diet – solid
 - Low residue for 6-8 weeks after in case of colostomy
 - All time in case ileostomy
3. Observe and record stoma condition
 - Beefy red and swelling (1st day)
 - Pink or red = ok
 - Notify HCP if stoma become black, dark blue, purple, pale = ↓ O2 = necrosis
4. Observe and record drainage from stoma
 - 1 -2 day - Serosanguineous = expected
 - 3 – 6 Days = fecal drainage begins.

- Irrigation colostomy = 1 week after surgery (never in ileostomy)
- 5. Promote positive adjustment to ostomy
- Encourage clients to look at the stoma.
- Encourage early participation in ostomy care
- Reinforce positive aspects of colostomy.

Colostomy Care = LPN

1. Wear clean gloves.
 2. Remove all appliances = proper fit of appliances = 1/8-inch space around the stoma.
 3. Assess the stoma condition.
 4. Clean stoma with mild soap and water and dry.
 5. Apply stoma adhesive paste to the skin = **skin sealant under all tapes.**
 6. Attach the ostomy drainage bag to the abdomen.
 7. Monitor the drainage.
1. Empty pouch 1/3 – ½ full
 2. Use pouch deodorant = Food to ↓ bad odor of colostomy:
 - Buttermilk
 - Parsley
 - Yogurt

Note: La bolsa de irrigation tiene que estar a nivel del shoulder, nunca +arriba porque va muy rápido, ni + abajo porque va lento.

Colostomy irrigation steps (toilet)

1. Purpose to stimulate emptying of colon at schedule time
2. 5 – 7 days post – op. beginning
3. 500 – 700 ml of tepid water
4. Special irrigation sleeve and cone are used
5. Insert catheter 8 cm
6. Hang irrigation container at shoulder high
7. Stop procedure if client complain of cramp or if water is forcefully returning around cone.
8. Time required = 15 – 45 min
9. When possible, the client should be sitting on the toilet

Diverticulitis

Diverticulosis	Diverticulitis
- Outpunch - Diet = high fiber to prevent diverticulitis	- Inflammation and infection of diverticula. - NPO/ no fiber no seeds

Assessment

- Pain LLQ that increases with coughing, straining or lifting.
- Fiver, flatulence, abdominal distention.
- Palpable tender rectal mass
- Blood in the stools

Intervention

1. IV antibiotic = Metronidazole/Ciprofloxacin
2. Antispasmodic = dicyclomine
3. NPO status = bowels rest
4. IV fluid
5. Surgical Intervention in case of perforation
6. Avoid seeds (semillas) -okra (kimkombo)